

Reviewer Instructions

Evaluating LLM Generated Peer Reviews of Academic Papers

About this study: When researchers submit a paper, they make specific claims of novelty in the introduction. A critical question in peer review is whether the methods section actually delivers on those claims. This study developed a technique to generate reviews using LLM and evaluates whether the LLM can automatically assess the alignment between a paper's stated_novelty claims and its methods section and whether that automated assessment agrees with the judgment of expert human reviewers who evaluated the same paper.

Your ratings will help us determine how well this automated claim-verification approach captures the same concerns that human reviewers independently identified across four categories of concerns: Novelty Concern, Methodology Issues, Clarity Issues, and Others

Data Sources:

- **Human Reviews** are from OpenReview.net — expert peer reviews submitted during the academic conference evaluation process. An LLM organizes these human reviews into four standardized categories of concerns (Novelty Concern, Methodology Issues, Clarity Issues, and Others)
- **Papers Under Review:** The same papers evaluated by human reviewers on OpenReview.net were used as input for the LLM-based assessment
- Review A & Review B: Generated independently by LLM
- Both LLM reviews are independent assessments

You are given an Excel workbook called paper_reviews.xlsx. There are 20 papers which are present in one tab each. Each paper has one human review and two independent LLM reviews.

- Column A — Human Review: a summary of concerns raised by human expert reviewers, organized into four categories — Novelty Concern, Methodology Issues, Clarity Issues, and Others

- Column B — Review A: an LLM-generated assessment of whether the paper's methods section substantiates the novelty claims made in the introduction, organized into the same four categories
- Column C — Review B: another LLM-generated review summary, organized into the same four categories

Your task

For each paper please follow the following procedure to evaluate each LLM review independently. For each LLM review ask yourself this question: does this review raise the same concerns as the human review across the four categories?

1. Read the Human Review carefully — note what concerns are raised in each of the four categories
2. Read Review A and rate the human review vs Review A on scale 1-5
3. Read Review B and rate the human review vs Review B on scale 1-5 independently

Rubrics:

Rating	Description
1	The LLM review raises completely different concerns from the human review — no meaningful overlap across any category
2	The LLM assessment touches on some of the same concerns but misses the main issues the human raised
3	The LLM review captures roughly half of the concerns the human reviewer identified across the four categories
4	The LLM assessment captures most of the same concerns with only minor gaps

5	The LLM assessment identifies the same concerns as the human review with no meaningful gaps
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Guidelines

- Please rate each paper independently. Do not compare papers against one another.
- If a review references work or terminology you are not familiar with, focus on whether the topics, methods, and claims it discusses match the paper described in the human review.
- We are evaluating the LLM reviews, not you. There are no wrong answers.
- Please do not use any AI tools (ChatGPT, Claude, etc.) to help generate your ratings. We need your independent expert judgement.

Time and logistics

We expect each paper to take roughly 4–6 minutes. The full set is 20 papers, so plan for around 1–1.5 hours total. You do not need to finish in one sitting — take breaks between papers.

When you are done, please return the completed response spreadsheet.

Questions and contact

If anything in these instructions is unclear, or if a paper tab appears to be missing data, please pause and contact me at rb194@students.uwf.edu before continuing. Thank you for your time and careful attention — your ratings are what makes this study possible.